SOFTWARE ENGINEERS 4 GREEN DEAL

CAPSTONE PROJECT

Team SuSoft

SusAF + Scrum Hackathon Report

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1. Hackathon Challenge Overview

This hackathon was organized by LUT University as part of the Software Engineering for the Green Deal (SE4GD) program. The challenge aimed to address the lack of awareness and knowledge among software development teams—including developers, product owners, Scrum Masters, programmers, designers, and other practitioners—regarding how to integrate sustainability metrics in the design and development of software products and services.

Sustainability is often overlooked in Agile workflows, with teams primarily focusing on business and technical requirements. The Sustainability Awareness Framework (SusAF) offers a structured approach to assessing the sustainability impacts of IT products and services. However, integrating these impacts into agile workflows, particularly within Scrum, continues to be a challenge as teams struggle to translate sustainability assessments into actionable backlog items, track sustainability improvements, and prioritize them alongside functional and technical needs.

During this hackathon, our team explored how sustainability concerns can be systemically embedded into Agile development. Through this challenge, we developed a web tool that can aid teams to seamlessly integrate sustainability into Scrum by transforming how sustainability impacts and effects identified using SusAF can be seamlessly integrated into Scrum by transforming SusAF assessments into actionable tasks, tracking sustainability decisions throughout the development cycle, and defining metrics for continuous improvement.

1.1 Problem Statement

In modern software development, sustainability considerations are rarely integrated into Agile workflows, making it difficult for teams to assess and improve the sustainability impact of their software products. While the Sustainability Awareness Framework (SusAF) provides a structured methodology for evaluating these impacts, many teams lack the knowledge and tools to translate sustainability insights into their Scrum-based development processes effectively.

1.2 Key Challenges

1. **From Awareness to Action:** Effectively translating sustainability impacts identified using SusAF into actionable SCRUM backlog items. Also incorporating sustainability concerns into user stories, product roadmaps, and feature requirements.

2. **Sustainability in Scrum Workflows:** Ensure teams have sustainability considerations actively included in sprint planning and sprint execution. Likewise, ensuring sustainability-related backlog items are prioritized alongside traditional business and technical requirements.

3. **Tracking Sustainability in Agile Development:** Ensuring software teams track sustainability decisions and their influence on design choices throughout the development cycle. Additionally, providing metrics to measure and review sustainability improvements in Sprint Reviews and Retrospectives.

4. **Sustainable Impact in Agile Metrics:** Provide strategies to be used by teams to continuously monitor and improve sustainability efforts in Scrum. Understanding how sustainability can be incorporated into the team's Definition of Done or other Agile best practices.

1.3 Identifying the Problem/Development Process

To ensure that our solution addressed the challenge, we followed a methodology to identify the core problem and its implications within software development SCRUM workflows. Our development approach followed the Agile and Scrum methodologies while integrating sustainability considerations at every step. The key steps include the following:

1.4 Understanding the Challenge and Conducting Initial Research

We started by reading and analyzing the hackathon challenge and problem statement. Each team member independently reviewed the challenge description, and the materials provided to ensure a clear understanding of the objectives.

To deepen our knowledge, everyone conducted individual research on related topics and shared relevant sources. Through this collective research effort, we developed a shared knowledge base that informed our discussions and guided our next steps.

1.5 Collaborative Brainstorming

After gathering initial research insights, we moved into a brainstorming session where each team member provided their perspective on the problem based on their research. During this phase:

- We encouraged open discussions on the main difficulties software teams face when trying to integrate sustainability.
- We listed various pain points, including the lack of sustainability prioritization, missing tracking mechanisms, and unclear guidelines.
- We mapped out connections between different challenges, which helped us visualize the problem landscape and refine our understanding of key obstacles.

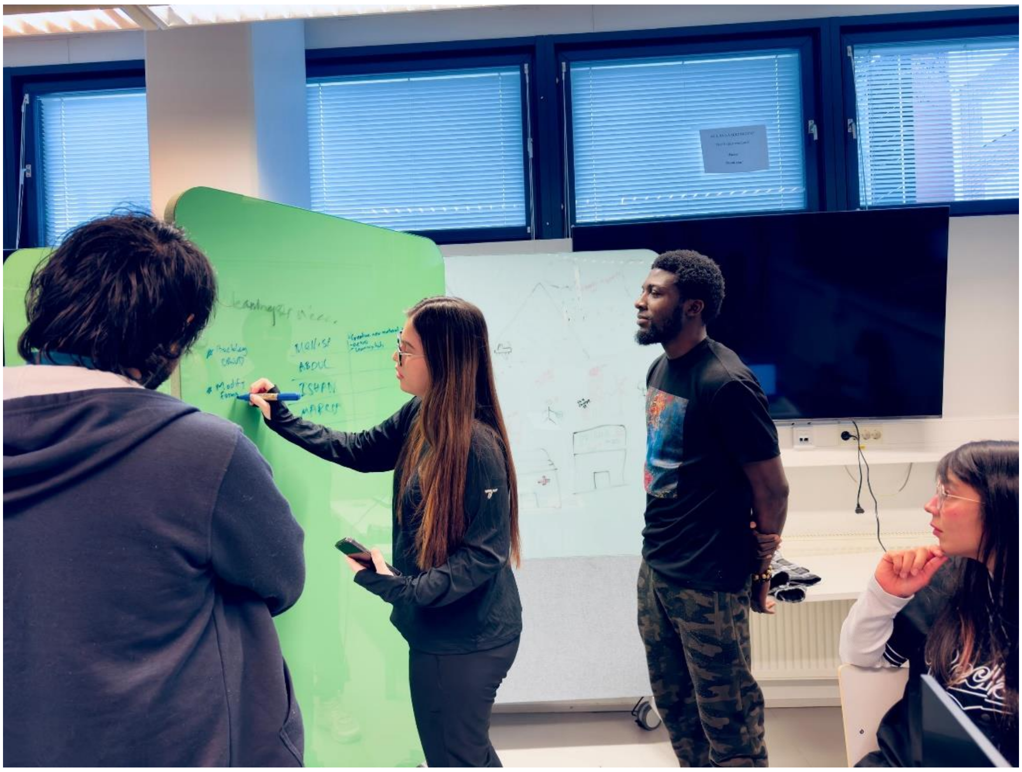

Figure 1: Collaborative Brainstorming

1.6 Problem Identification & Framing

On completing the brainstorming session, we synthesized our findings into a well-defined problem statement:

“How can software development teams effectively translate sustainability concerns into actionable Scrum backlog items, prioritize them alongside other requirements, and track their impact throughout the development cycle?"

Through the process of refining and validating, we ensured that we were focused on the challenge's theme and addressing a tangible and meaningful issue within Agile sustainability integration.

1.7 Exploring Potential Solutions & Decision-Making

With a clear problem statement established, we brainstormed and explored viable solutions to approach our problem statement. A few of the many options we had to explore include:

i. A **checklist-based approach** where sustainability criteria are manually integrated into Scrum.

ii. A **plugin for Agile tools** (e.g., Jira, Trello) to automate sustainability requirements with the help of an AI assistant.

iii. A **web-based platform** that provides structured guidance and integrates with Agile workflows.

After carefully discussing these options while considering the pros and cons of each solution, we decided to focus on the web-based tool that provides structured sustainability integration into Scrum workflows, incorporating elements of education, task tracking, and automated guidance through an existing SusAF web-based tool. This also reduces the time for developing a solution from scratch, considering the time interval allocated to complete the challenge.

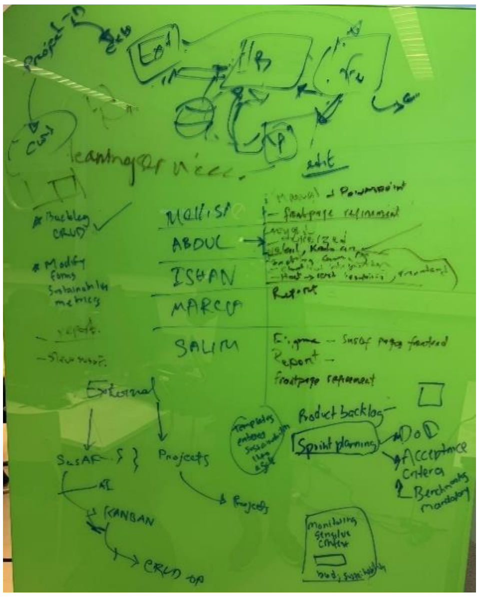

Figure 2: Exploring Potential Solutions

1.8 Sketches & Wireframing:

To bring our ideas to life, we used Figma to create the UI/UX wireframes for our solution. Additionally, the key user journeys and interface flow were defined in the initial wireframe before implementing the full prototype. The sketch and implementation procedures in Figma are shown in the images below:

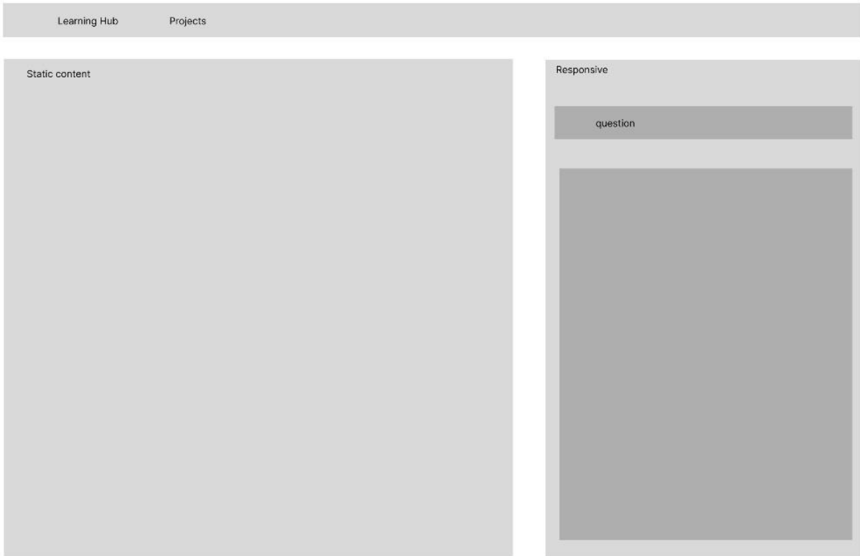

Figure 3: Wireframe #1

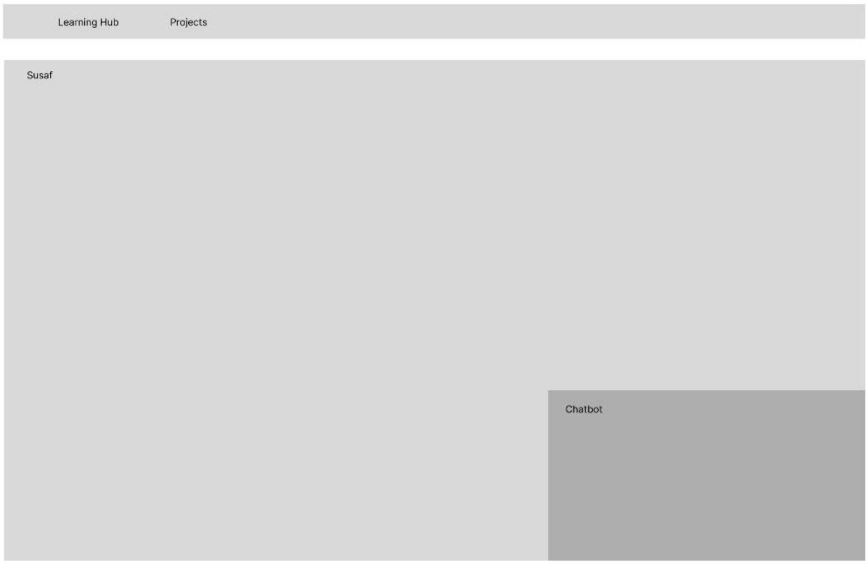

Figure 4: Wireframe #2

1.9 Daily Stand-ups & Agile Rituals

We held daily stand-up meetings to track progress and adjust tasks, conducted Sprint Planning to incorporate sustainability criteria into backlog items, focused on iterative development during Sprint Execution to refine features, and assessed sustainability tracking and usability in Sprint Reviews and Retrospectives.

2. Our Solution

Our solution is a **web-based** tool that integrates sustainability considerations into Scrum workflows. We have decided to use the concept of common tools such as Jira and Kanban board to increase acceptability, as many companies are used to those commercial tools.

2.1 SUSAF Analysis of our Solution

2.1.1 Identified possible sustainability effects

To conduct the SUSAF analysis of our solution, we identified all possible effects of its intended features. Then we selected the features with high impact and likelihood. They are presented in the table below

Features	Dimension	Effect
Integration with Agile tool	Social	Raises awareness about sustainability in software development, fostering a culture of responsible engineering.
	Economic	Reduces long-term costs by educating and ensuring teams on sustainable practices that improve efficiency.
	Environmental	Encourages eco-friendly coding and design choices, leading to lower energy consumption.
	Environmental	Leverages on Large Language models for this integration and these are known for energy consumption
	Social/Economic	Requires time investment for training, which may slow down development cycles.
AI Chatbot Integration	Social	Provides guidance on sustainability, making it easier for teams to integrate eco-conscious decisions.

	Environmental	Relies on LLM model to respond to inquiries. These are energy-consuming.
Sprint Execution and Tracking	Economic	Improves efficiency by identifying sustainability obstacles early.
Sustainability Acceptance Criteria	Economic	Meeting sustainability benchmarks may require additional resources and increase costs.
	Social	Helps businesses maintain compliance with sustainability standards.
	Environmental	Guarantees that software development aligns with eco-friendly principles

NB: Text Highlighted in red signifies negative impact

2.1.2 Identified Opportunities

Using the SusAF webtool the following opportunities were identified:

a. The effect chain highlights the importance of integrating eco-conscious decisions in software development, which can foster a culture of sustainability that improves team morale and attracts environmentally conscious clients.
b. By educating teams on sustainable practices and reducing long-term costs, companies can leverage these practices as a competitive advantage, potentially leading to increased market share and customer loyalty.

2.1.3 Identified Threats

a. The reliance on Large Language Models (LLMs) introduces a significant energy consumption variable, which could undermine environmental goals if not properly managed.
b. Meeting sustainability benchmarks may require upfront investments and additional resources, which can pose a financial burden for some organizations, particularly small businesses.

2.1.4 Recommendations

a. Leverage local hosting of LLM model for backlog generation and sustainability chatbot to reduce their energy consumption while balancing their usage for eco-friendly coding and design decisions.

b. Implement Retrospective to consistently evaluate the team's progress on adopting sustainability in their workflow.
c. Consider phased investments in sustainable practices that allow organizations to gradually adapt and fund their sustainability initiatives without significant financial strain.
d. Implement support for learning of sustainability at developers' choice of pace to avoid overwhelming developers.

2.2 Key Features

Based on the objective of the tool and the recommendation of Susaf Analysis, the web tool consists of six (6) main feature screens, each serving a specific purpose in ensuring sustainability is effectively embedded into the software development process:

1. **Learning Hub:** We have considered our stakeholders to be users without prior knowledge of sustainability and would require some basic understanding of sustainability before engaging the tool. This is a major reason the learning hub has been designed to be the entry point where users are introduced to sustainability concepts. It provides educational content on sustainability in software development, ensuring that all team members have foundational knowledge of sustainability, its dimensions, and how they relate to sustainable software development before proceeding to make use of the SusAF tool.

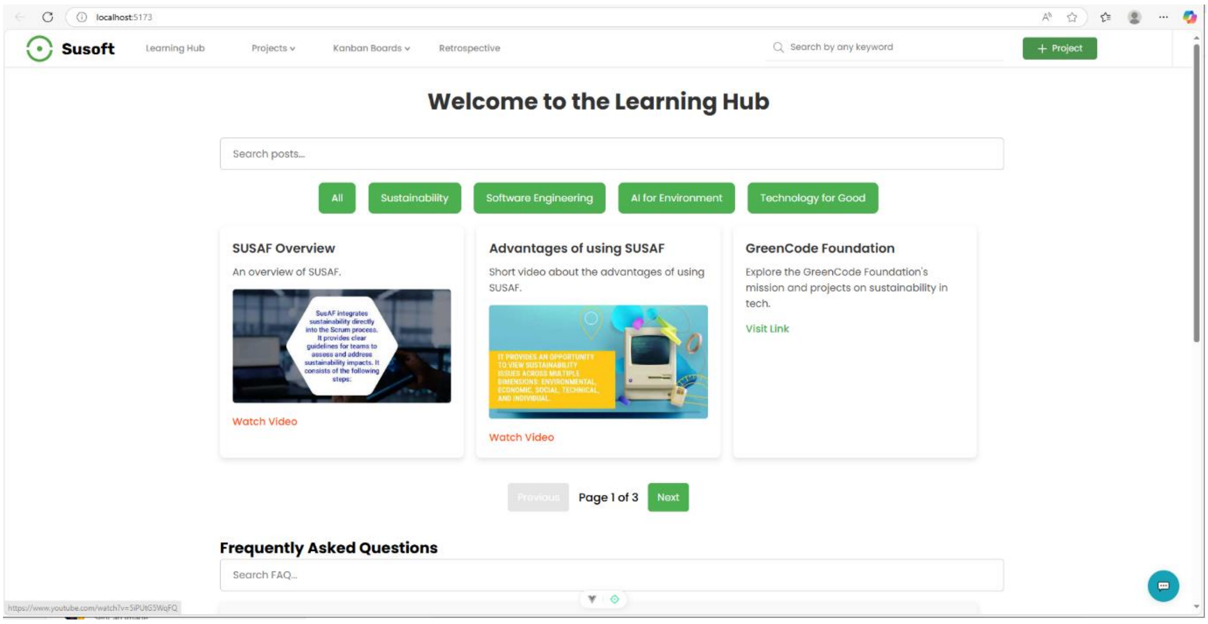

Figure 5: Learning Hub

2. **AI Sustainability Chatbot:** Our web tool hosts the SusAF tool, and it is integrated with an AI-powered chatbot that assists users in understanding sustainability-related aspects. This chatbot guides the team across all the development stages in the agile development process. Additionally, it clarifies the impact of chain effects across all three (3) dimensions of sustainability and helps users navigate the SusAF framework while having sustainability concerns in mind. This aims to make it easier to identify sustainability concerns relevant to their project at every stage of the development process.

   It is important to know that we have considered the impact of the energy consumption of AI systems before taking the design decision of introducing a chatbot into our solution. However, to mitigate and reduce the consumption of this AI solution, we decided to host the chatbot locally, which reduces the amount of energy consumed through different queries that are made on the chatbot. This has proven to be a sustainable approach to ensure system efficiency while considering the impact of the solution on the environment. Below is a sample conversation with the sustainability chatbot:

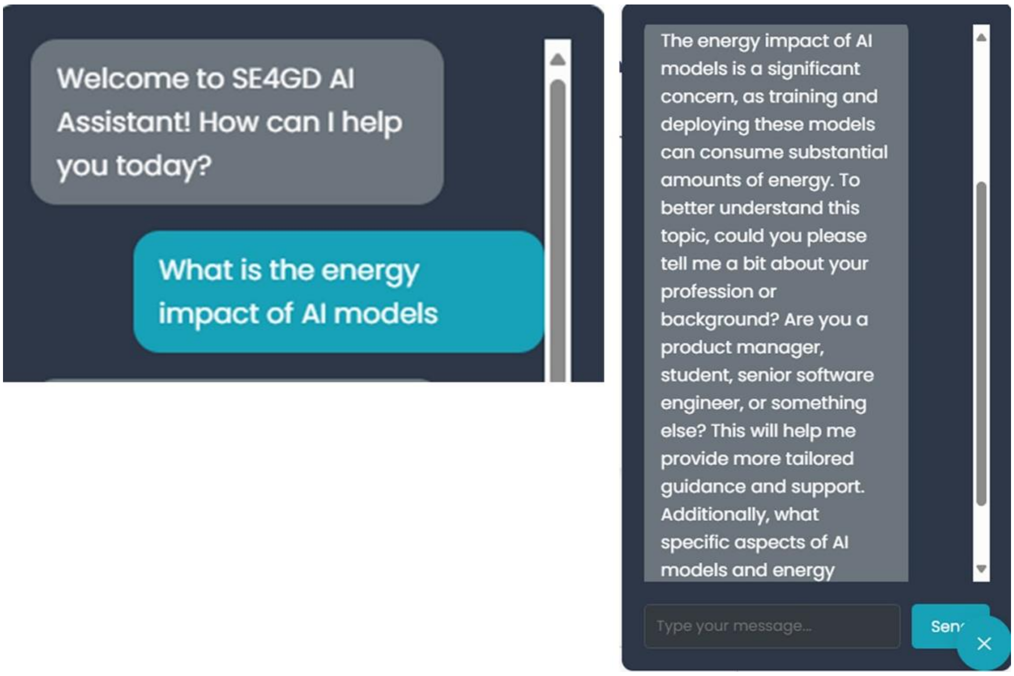

Figure 6: Sustainability Chatbot

3. **Backlogs Generation:** The Backlogs screen is a key feature of our system that allows teams to systematically integrate sustainability considerations into Agile development. It serves as a central hub for creating, managing, and prioritizing sustainability-related backlog items alongside traditional business and technical tasks. Some of the key insights from our solution include:
4. **Creating Sustainability Backlog Items:** Teams can enter a project ID from the SusAF web tool to create a new project on Susoft. Susoft collects the recommendations from the Susaf web tool and, leveraging a locally hosted model, generates backlog items from the recommendations.

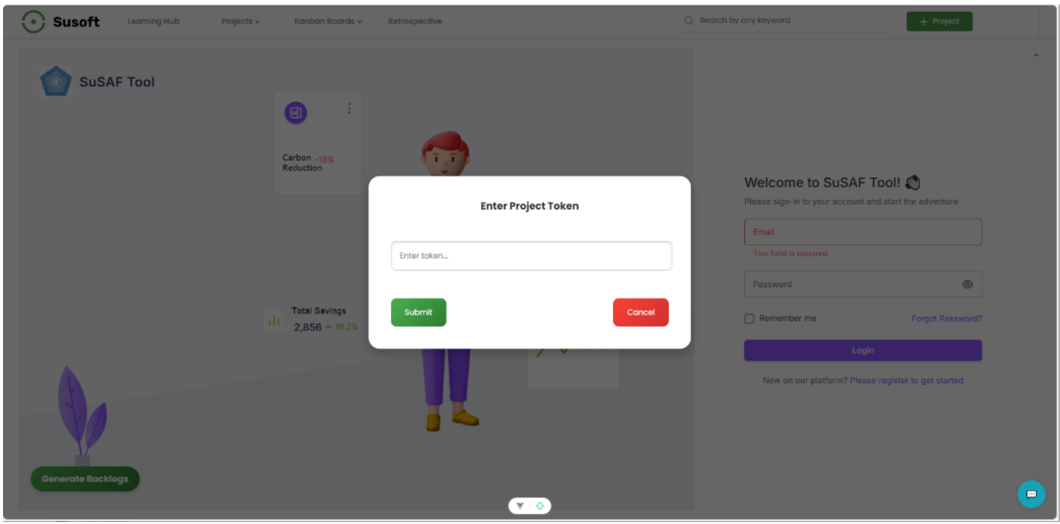

Figure 7: Creating a project and generating backlogs

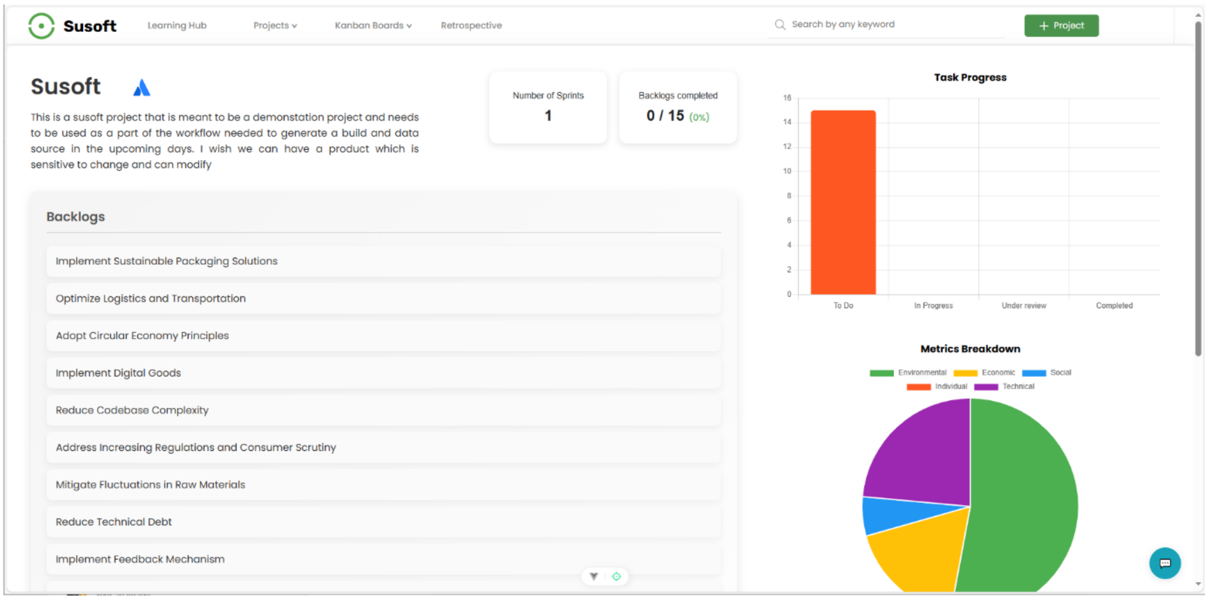

Figure 8: Backlogs page

5. **Prioritization & Sprint Planning Integration**: Sustainability concerns are prioritized based on impact and integrated into sprint planning, ensuring they are actively addressed alongside other development tasks.
6. **Alignment with Scrum Workflows:** Backlog items are categorized by Scrum phases—development, testing, deployment, and monitoring—to keep sustainability an ongoing focus.
7. **Tracking and Refinement:** Items evolve through sprint reviews and retrospectives, allowing teams to track progress, measure impact, and refine sustainability strategies. This feature embeds eco-conscious decision-making into Agile workflows, making sustainability a core aspect of software development.

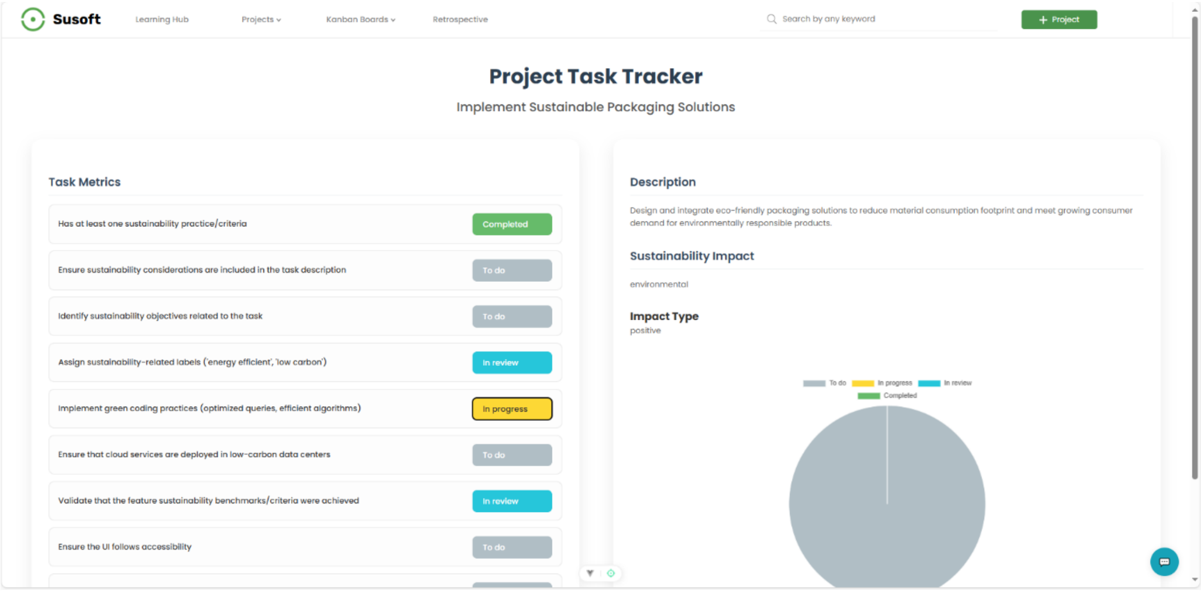

Figure 9: Task Tracking page

8. **Sprint Execution:** In this phase, teams continuously track and document sustainability challenges, ensuring timely identification and resolution. Teams log sustainability-related obstacles as they arise, allowing for quick adjustments. Also, developers, Scrum Masters, and product owners collaborate to address challenges and refine sustainability strategies. Insights gathered during the sprint help refine backlog items, ensuring sustainability considerations evolve with development needs. This process ensures sustainability remains a core focus throughout the sprint, fostering a proactive approach to eco-conscious software development.

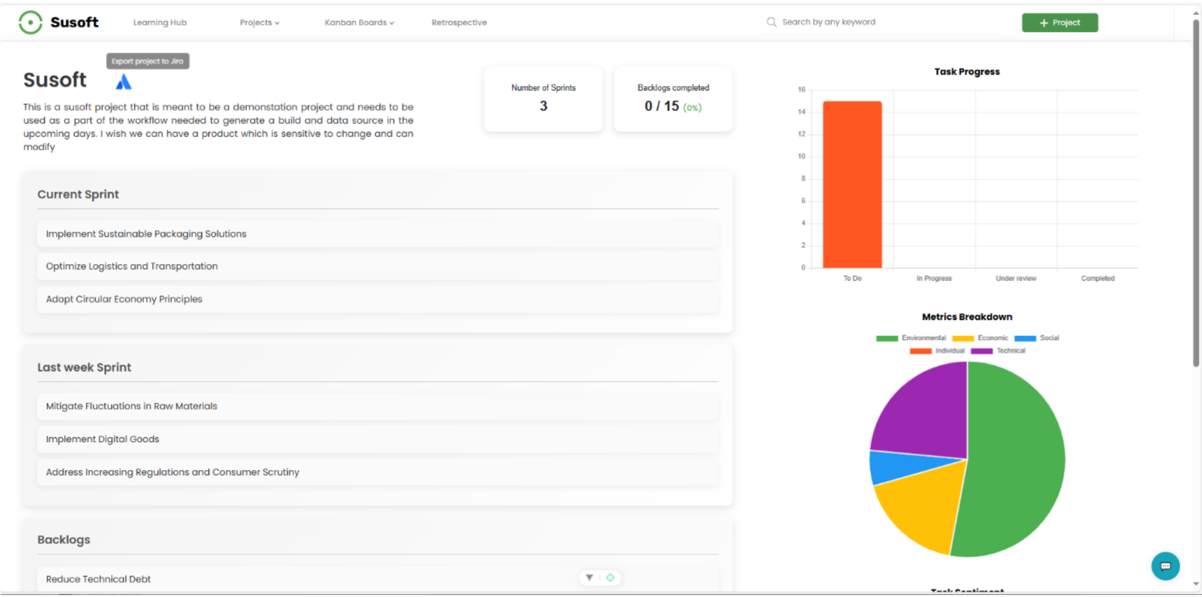

Figure 8: Sprint Execution

9. **The Kanban Board:** The fourth screen is a Kanban-style board that provides a visual workflow for managing tasks, ensuring sustainability-related items are integrated alongside other development priorities. Some of the notable features include:
    i. **Task Management:** Teams can track and update sustainability tasks in real-time, maintaining visibility across all development phases.
    ii. **Seamless Integration:** Sustainability tasks move through the workflow like any other backlog item, ensuring they remain a core focus.
    iii. **Progress Tracking:** Teams can monitor sustainability efforts alongside business and technical tasks, reinforcing accountability and continuous improvement.

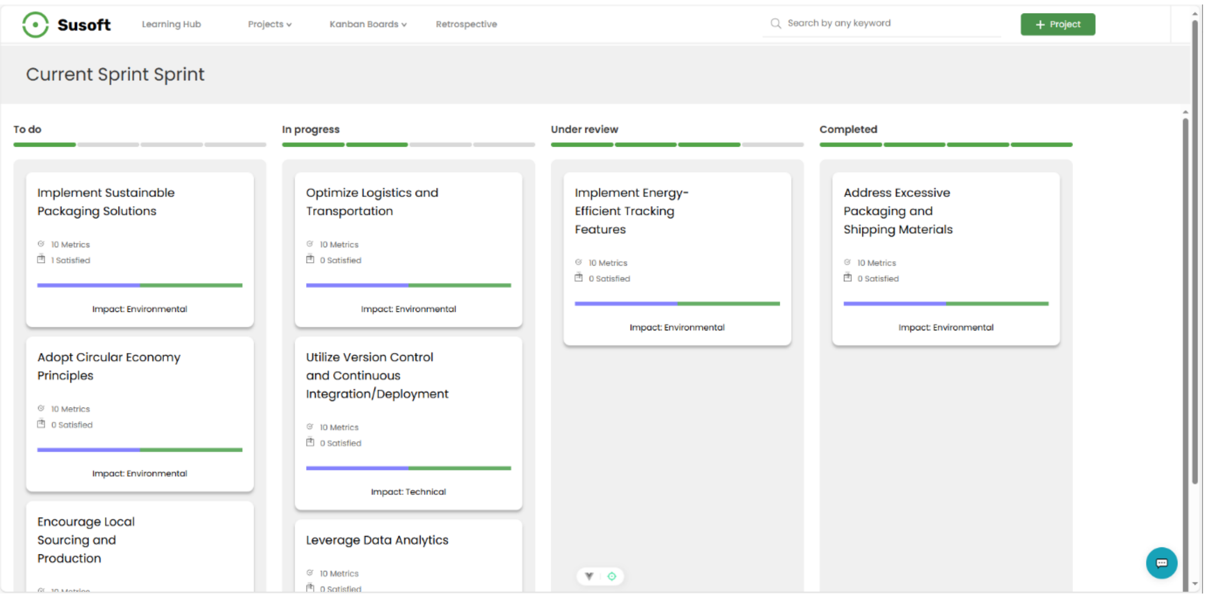

Figure 11: Kanban board

This feature ensures sustainability remains an active part of Agile workflows, enhancing visibility and execution.

10. **The Retrospective:** This screen enables teams to reflect on sustainability efforts during each sprint and identify areas for improvement.
    i. **Sustainability Assessment:** Teams review the effectiveness of implemented sustainability initiatives.
    ii. **Actionable Insights:** Identifies successes and areas needing improvement for future sprints.
    iii. **Continuous Improvement:** Ensures sustainability remains a key focus in Agile development cycles.

This feature helps teams refine their approach, making sustainability a continuous and measurable practice.

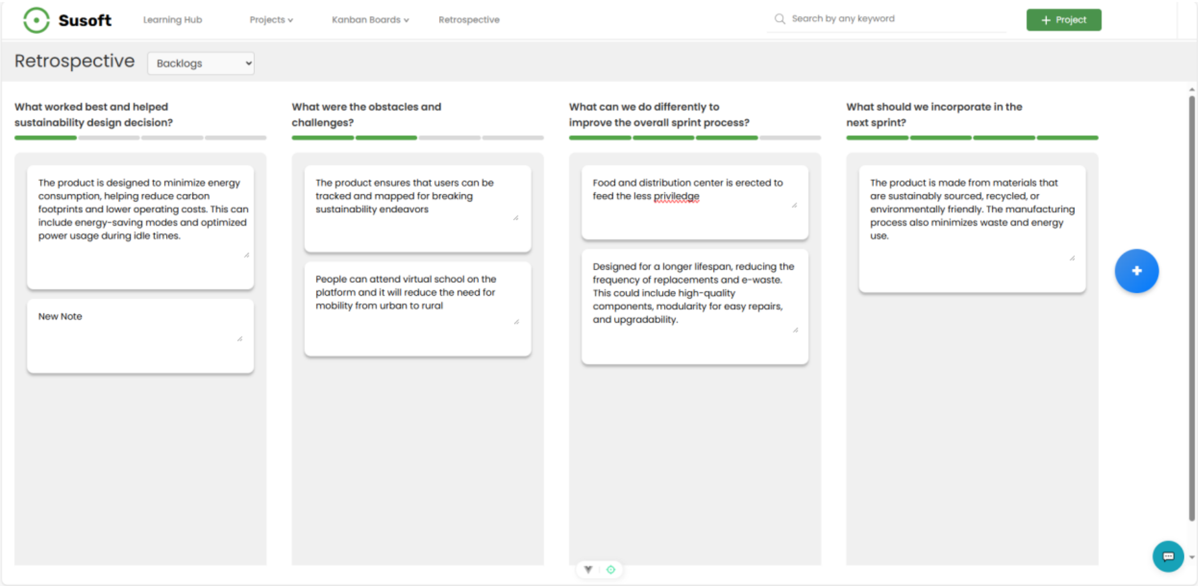

Figure 12: Retrospective

11. **Task Details & Sustainability Acceptance Criteria**: The Task Details screen provides insights into individual tasks, ensuring sustainability is embedded in development. Some of the key features of this screen include:
    i. **Sustainability Acceptance Criteria:** Defines key sustainability factors that must be met for task completion.
    ii. **SusAF Integration:** Displays outputs from SusAF, helping teams align tasks with sustainability principles.
    iii. **Agile Best Practices:** Ensures sustainability remains a core consideration throughout the development lifecycle.

This feature reinforces accountability, guiding teams toward sustainable and responsible software development.

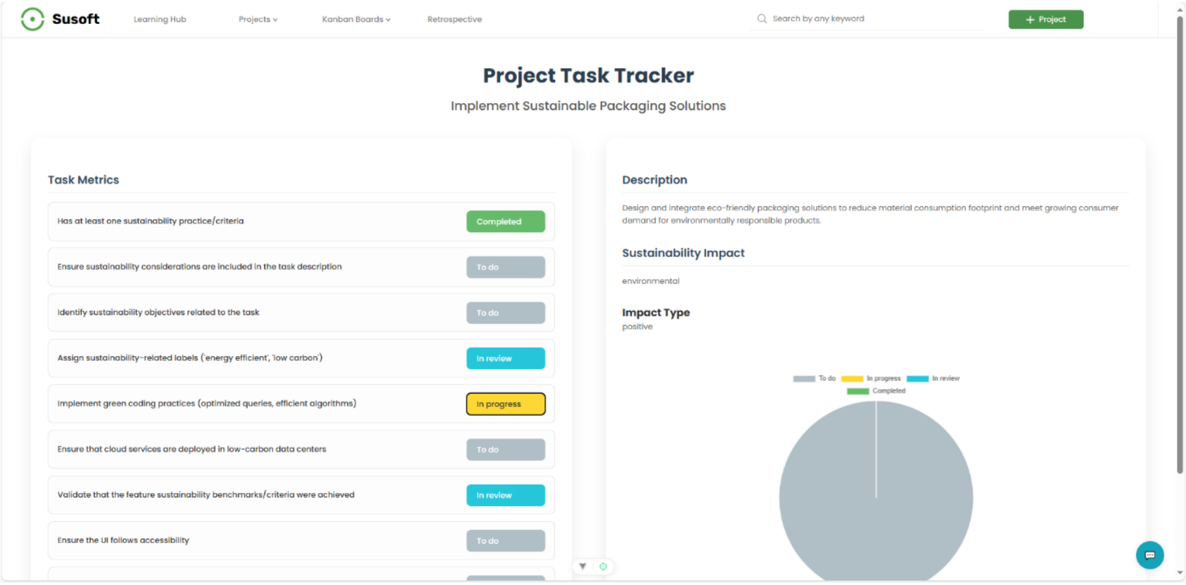

Figure 13: Project Task Tracker

3. Technology Stack/Implementation

For solution implementation, we strategically selected a technology stack that aligned with our team's expertise and seamlessly integrated with the existing **SusAF web tool**. Each component was chosen to optimize development efficiency and ensure smooth integration across various stages of the project.

1. **UI/UX Design:** We used **Figma** to transform initial sketches on the drawing board into a functional prototype, streamlining the development process. The wireframes provided a clear visual direction, ensuring a seamless transition from design to implementation while offering a high-level overview of the final solution.
2. **Front-end development:** We selected **Vue.js** due to its flexibility, performance, and ability to create a dynamic, user-friendly interface. Vue's reactive data binding and component-based architecture allowed us to efficiently build an interactive UI that seamlessly integrates sustainability features into Agile workflows.

   Additionally, Vue.js offers:

   i. Lightweight and Fast Performance: Ensures a smooth user experience with minimal load times.

   ii. Modular Structure: Simplifies development, making it easier to manage and scale components.

   iii. Seamless State Management: Enables efficient handling of sustainability-related backlog items and real-time updates.

iv. Ease of Integration: Works well with existing SusAF web tools, ensuring a cohesive solution.

By leveraging Vue.js, we ensured our application delivers a highly intuitive and responsive user experience, enhancing usability for Agile teams integrating sustainability into their workflows.

3. **Backend development:** We implemented this aspect with Node.js, leveraging its non-blocking, event-driven architecture to ensure high performance and scalability. Node.js was chosen for its efficiency in handling asynchronous operations, making it ideal for real-time updates and seamless communication between the front end and database.
   We utilized Node Version Manager (NVM) to manage different Node.js versions efficiently, ensuring compatibility across various development environments. Additionally, Node Package Manager (NPM) was used to manage dependencies, streamline package installations, and facilitate modular development. Another important feature we considered was its seamless integration with the Vue.js front end and database, ensuring a cohesive system architecture.
   By leveraging **Node.js with NVM and NPM**, we built a robust backend capable of efficiently managing sustainability workflows within Agile development.
4. **Database:** We implemented PostgreSQL as the database for its reliability, scalability, and structured data management, ensuring efficient storage and retrieval of sustainability-related backlog items, sprint data, and user interactions.
5. **API Integration:** To seamlessly incorporate the existing SusAF web tool within our solution while optimizing development time, we integrated it using an **IFrame**. This approach allowed for a lightweight and efficient embedding of SusAF, ensuring smooth interaction without requiring extensive redevelopment. By leveraging an **IFrame**, we maintained full functionality and accessibility, enabling teams to utilize SusAF insights directly within our system.
6. **Deployment:** We used docker to containerize our solution. Later, we deployed the docker image in an Oracle Cloud compute instance. The deployment contains frontend, backend, and an AI backend microservice, where each of the concerns are separated, making them easy to scale without interrupting other services. We used shared cloud machine to prevent resource waste, ensuring as much technical sustainability as possible in every step of the deployment process.
7. **Version Control:** We used Git for version control and maintained an open-source GitHub repository, enabling seamless collaboration among team members. With different

contributors working on various aspects of the project, Git ensured efficient branching, merging, and conflict resolution, streamlining development, and maintaining code integrity.

3.1 System Architecture

Our system is designed to seamlessly integrate sustainability considerations into Agile workflows. It comprises the following key components and a high-level architecture is shown in Figure 10.

- **User Authentication:** Secure login system to our tool which serves as a link to the SusAF web tool.
- **Backlog Integration Module:** Converts SusAF sustainability factors into actionable backlog items.
- **Sprint Planning Dashboard:** Enables teams to prioritize sustainability concerns effectively.
- **Tracking and Metrics Dashboard:** Monitors sustainability efforts and progress across sprints.
- **Reporting & Documentation:** Generates reports and provides exportable templates for Agile teams.http://129.213.86.120:3000/

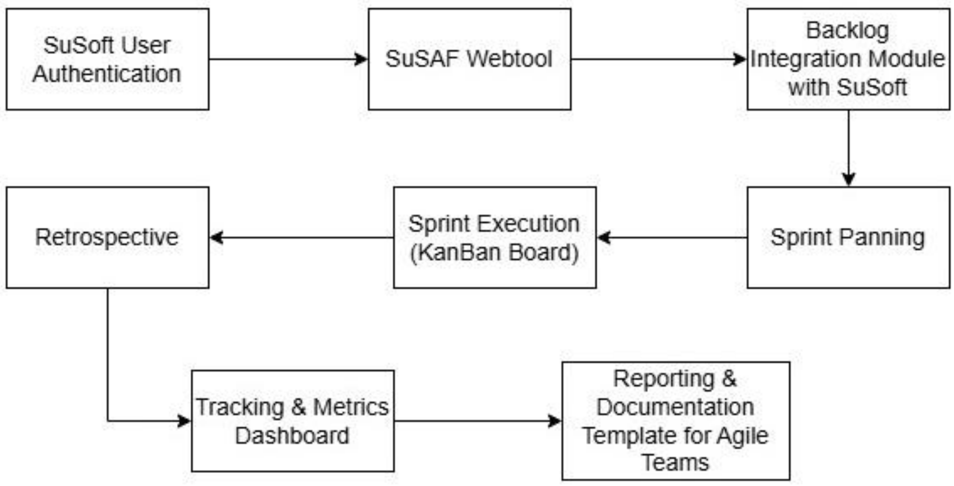

Figure 14: System Architecture

4. Shift in Mindset

While SuSoft offers a functional platform to integrate sustainability into Scrum workflows, its deeper impact lies in promoting a fundamental shift in mindset among software teams.

This transition mirrors the transformation that occurred when the Agile Manifesto first emerged, a shift not merely in methodology but in how developers, teams, and organizations thought about building software.

Incorporating sustainability into Agile is not about adding tasks or adjusting ceremonies alone. It requires rethinking the very definition of what it means to deliver "value." The traditional focus on velocity, deliverables, and deadlines must expand to include long-term environmental, social, technical, and economic impacts. Teams must transition from being feature-driven to becoming impact-aware.

Figure 15: Shift in Mindset

4.1 From Awareness to Culture

Behavioral change within development teams does not occur overnight. It must be designed and cultivated across multiple layers — through daily rituals, visible cues, emotional motivation, and team-wide accountability. Our approach is grounded in the idea that lasting transformation happens across five stages:

Visibility → Awareness → Action → Habit → Culture

SuSoft is designed to activate this transition by making sustainability not only visible but also engaging and personally meaningful.

4.2 Building Engagement Through Cultural Cues

To support this cultural evolution, SuSoft incorporates a range of awareness-building and identity-reinforcing tools that go beyond traditional software solutions. These cultural assets were created to encourage teams to internalize sustainability values and celebrate their progress. Examples include:

Visual Campaigns: A series of bold, minimalist posters carrying slogans such as “**Sprint Like the Future Depends on It**” and “**Because the Planet Can’t Wait for the Next Sprint**” to build emotional resonance and team alignment.

Tangible Reinforcement: Stickers, pins, badges, and woven patches featuring SuSoft branding and motivational messages, serving as daily reminders of the mission. Limited-edition "Sprint Champion" pins and badges to reward team members or groups who actively embrace sustainable sprint practices.

Figure 16: Branding Ideas

Role-Based Action Figures: Custom-designed action figures representing key Scrum roles (Product Owner, Scrum Master, Developer) as collectible items, bringing identity and play into the culture shift. These action figure trend have been a great trend in 2025 for community building. These kinds of social media trends aim to engage gene-z employees into the campaign.

Figure 17: Role-Based Action Figures

Recognition Mechanisms: Limited-edition "Sprint Champion" pins and badges to reward team members or groups who actively embrace sustainable sprint practices.

Learning Infrastructure: A built-in learning hub that encourages ongoing reflection, exploration, and upskilling around sustainability topics relevant to software teams.

These elements are not ornamental — they are behavioral catalysts, rooted in principles of nudging, habit formation, and organizational psychology.

4.3 A New Definition of Agile Success

Success in Agile can no longer be defined solely by speed and delivery. In an age of ecological, social, and ethical urgency, success must also mean building responsibly, iterating consciously, and designing systems that are not just scalable, but sustainable.

SuSoft aims to pioneer this evolution - from delivery-focused teams to sustainability-conscious cultures.

5. Conclusion

This hackathon challenged us to explore the intersection of sustainability and Agile development. By leveraging SusAF, we successfully built a web tool that helps Scrum teams integrate sustainability into their workflows. The project not only highlights the importance of sustainability in software development but also provides practical solutions for Agile teams to adopt.

6. References and Appendices

- A fully functional web tool integrating sustainability into Scrum workflows can be found here (SuSoft).
- Presentation slides summarizing project outcomes (Link here)
- GitHub Repository: GitHub Repo
- User Manual can be found here (Link here)
- External resources for sustainability enthusiasts (Link here)